Exercises

Problem 2.1

Find the norm of the pure qubit state $|\psi\rangle = \frac{1}{3}|0\rangle + \frac{i}{\sqrt{3}}|1\rangle$ and the corresponding normalised density matrix. Locate the state on the Bloch sphere.

Problem 2.2

Find the overlap $\langle\psi|\phi\rangle$ of the two qubit states $|\psi\rangle = \sqrt{\frac{1}{3}}|0\rangle + i\sqrt{\frac{2}{3}}|1\rangle$ and $|\phi\rangle = \sqrt{\frac{1}{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$. Locate both states on the Bloch sphere.

Problem 2.3

Consider the pure qubit state $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle$. Find the corresponding density matrix $\hat{\rho}$ and find the Bloch vector $\mathbf{u}$ such that $\hat{\rho} = \frac{1}{2}(\mathbb{1} + \mathbf{u}\cdot\hat{\boldsymbol{\sigma}})$.

Problem 2.4

Find the matrix representation of the qubit operators $\hat{\sigma}_\pm = \frac{1}{2}(\hat{\sigma}_x \pm i\hat{\sigma}_y)$ in the standard basis .

Problem 2.5

A qubit is in the state $|\psi\rangle = \cos(\theta)|0\rangle + i\sin(\theta)|1\rangle$ and a measurement of the Pauli operator $\hat{\sigma}_x$ is performed. What are the possible outcomes and what are their probabilities?

Problem 2.6

Prove Eq. (2.77). That is, for a qubit state of the form $\hat{\rho} = \frac{1}{2}(\mathbb{1} + \mathbf{u}\cdot\hat{\boldsymbol{\sigma}})$, prove that the components of the Bloch vector $\mathbf{u}$ are given by $u_k = \mathrm{Tr}[\hat{\rho}\hat{\sigma}_k]$, for $k = x, y, z$.

Problem 2.7

The joint state of two qubits is $|\psi\rangle_{AB} = \frac{1}{3}|00\rangle_{AB} + \frac{\sqrt{2}}{3}|01\rangle_{AB} + \frac{\sqrt{2}}{3}|10\rangle_{AB} + \frac{2}{3}|11\rangle_{AB}$. What is the reduced state of qubit A? Is the state $|\psi\rangle_{AB}$ entangled?

Problem 2.8

Illustrate the effect of the following operations on the Bloch sphere

$$\begin{aligned}
\hat{M}_1 &= \exp\left(-i\frac{\pi}{2}\hat{\sigma}_x\right), \\
\hat{M}_2 &= \exp\left(-i\frac{\pi}{4}\hat{\sigma}_y\right), \\
\hat{M}_3 &= \exp\left(-i\frac{\pi}{8}\hat{\sigma}_z\right), \\
\hat{M}_4 &= \exp\left(-i\frac{\pi}{4\sqrt{2}}(\hat{\sigma}_x + \hat{\sigma}_y)\right).
\end{aligned}$$

Problem 2.9

Consider the qubit state $\hat{\rho}$ with density matrix

$$\hat{\rho} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix},$$

in the computational basis. Find the the Bloch vector of the state. Show that $\hat{\rho}$ is not pure.

Problem 2.10

Show that the POVM element $\hat{E}_0$ as defined in Example 2.1 is positive semidefinite.

Problem 2.11: Positivity of reduced states

Show that the reduced state as defined in (2.66) is positive semidefinite (*hint:* show that $\langle\phi|\hat{\rho}_A|\phi\rangle) \geq 0$ for any $|\phi\rangle$).

Problem 2.12: Minimum-error state discrimination

You are given a large number of quantum systems with a promise that half of them are each in a pure quantum state $|\psi_0\rangle$ and the other half in $|\psi_1\rangle$. You are asked to sort them according to state.

If the overlap $|\langle\psi_0|\psi_1\rangle| = \delta$ is nonzero then the states are not orthogonal and therefore cannot be perfectly distinguished. Your task is therefore impossible. However, what is the best you can do?

We are looking for a measurement that can discriminate the states as well as possible. Consider a binary measurement and let $p(m|x)$ denote the probability that the measurement device outputs m when the state is $|\psi_x\rangle$, with $x, m \in \{0, 1\}$. The goal is to output $m = x$, successfully identifying the state, while $m \neq x$ is an error. The average error probability is

$$p_e = \frac{1}{2}\, p(1|0) + \frac{1}{2}\, p(0|1), \tag{2.102}$$

where the factors 1/2 arise because there are equal numbers of each state. The average success probability is $p_s = 1 - p_e$. We want to find the optimal measurement which minimises p_e (and thus maximises p_s).

(a) Argue that it is sufficient to consider qubit states and measurements, without loss of generality.

(b) Argue that the measurement can be defined in terms of a single operator $\hat{E}_0$ with $p(0|x) = \mathrm{Tr}[\hat{E}_0|\psi_x\rangle\langle\psi_x|]$.

(c) Show that the measurement is optimal when $\hat{E}_0$ is a projector onto the positive-eigenvalue eigenvector of the operator $\frac{1}{2}(|\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1|)$.

(d) Show that the minimal error probability is $p_e = \frac{1}{2}(1 - \sqrt{1-\delta})$.

This fact is known as the *Helstrom bound*.

Problem 2.13: Unambiguous state discrimination

Consider the same situation as in Prob. 2.12, but with a slightly different task. You are again asked to sort the states but now *without making any errors* whatsoever. However, you are allowed to sometimes answer 'I don't know'. Your goal is to sort while giving this inconclusive answer as rarely as possible.

Let $p(m|x)$ denote the probability that the measurement device outputs m when the state is $|\psi_x\rangle$, now with $x \in \{0, 1\}$ and $m \in \{0, 1, \varnothing\}$ where $\varnothing$ denotes

the inconclusive outcome. The goal is to minimise the inconclusive probability

$$p_{\varnothing} = \frac{1}{2}\, p(\varnothing|0) + \frac{1}{2}\, p(\varnothing|1), \tag{2.103}$$

with the condition that there are no errors $p(0|1) = p(1|0) = 0$.

(a) Argue that it is sufficient to consider qubit states and measurements, without loss of generality.

(b) Argue that the measurement cannot be projective and that it will be a POVM with three elements $\hat{E}_m$ such that $p(m|x) = \mathrm{Tr}[\hat{E}_m|\psi_x\rangle\langle\psi_x|]$.

(c) Show that the condition $p(1|0) = 0$ implies that $\hat{E}_1 = r_1|\psi_0^{\perp}\rangle\langle\psi_0^{\perp}|$, where $|\psi_0^{\perp}\rangle$ is a state orthogonal to $|\psi_0\rangle$ and $r_0 \geq 0$. Similarly, $\hat{E}_0 = r_0|\psi_1^{\perp}\rangle\langle\psi_1^{\perp}|$. *Hint: it is always possible to find a basis in which $|\psi_0\rangle$ is one of the elements. Similarly for $|\psi_1\rangle$.*

(d) Show that $p_{\varnothing} = 1 - \frac{1}{2}(r_0 + r_1)(1 - \delta^2)$.

(e) Show that $r_0 + r_1 \leq \frac{2}{1+\delta}$ for the measurement to be a valid POVM. *Hint: Consider the positivity of $\hat{E}_{\varnothing}$.*

(f) Construct an explicit POVM which achieves equality, $r_0 + r_1 = \frac{2}{1+\delta}$, thus proving that the optimal inconclusive probability is $p_{\varnothing} = \delta$.